

Abhishek Kumar

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RESEARCH INTERESTS

Computational Neuroscience, Neuromorphic Computing, Spiking Neural Network, Neural Modeling, Machine Learning, Brain-Inspired Artificial Intelligence

EDUCATION

ABV-Indian Institute of Information Technology and Management Gwalior Gwalior, MP
MS (Artificial Intelligence and Data Science) Aug. 2025 - Present

University of Delhi Delhi
Bachelor of Science, Mathematics (H) Nov. 2022 - Jun 2025

EXPERIENCE/PROJECTS

Winter Research Internship [Certificate](#) Dec. 2025 - Jan. 2026
Ion channel and Receptor biology lab, KSBS Indian Institute of Technology, Delhi

- Developed neuron simulations using **NEURON** to study ion-channel and spike dynamics.
- Implemented and modified biophysical neuron models using **hoc scripting** and explored ModelDB.
- Applied supervised learning models to molecular datasets for toxicity prediction, aligning with ML-driven drug discovery.

Aadiyogi Dec. 2022 - Jun. 2025
Team Captain Deshbandhu College, UoD

- Led the team in regional, state, and national-level yoga competitions and championships.
- Facilitated weekly practice sessions to discuss and master advanced Asanas, Pranayama & Meditation techniques.
- Acted as the primary liaison between team members, coaching staff, and competition officials.

PROJECTS

Ultra-Low-Power Edge Intelligence for Real-World Anomaly Detection Using SNN Ongoing
Spiking Neural Networks, Edge AI, Embedded Systems [Project info](#).

- Creating an event-driven anomaly detection framework to identify irregular patterns that indicate sewage blockages.
- Project currently under development.
- Implementation details and results will be shared upon completion

Smart Shield: Lightweight On-Device SMS Fraud Detection Using SNN Mar. 2026
[GitHub](#) Hacksecure'26 Hackathon, NIT-H

- Developing an Android application to detect fraudulent and spam SMS messages.
- Integrated a snnTorch-based classification model for real-time message analysis.
- Implemented preprocessing and inference pipeline for on-device ML deployment.
- Built during Hacksecure'26, a cybersecurity hackathon organized under ISEA Phase-III (MeitY, Govt. of India).

Air-Sense: Smart Air Quality Monitoring & Prediction System Mar. 2026
[GitHub](#) AI Synergy Hackathon' 26, IIIT Gwalior

- Developed a ML-based system to monitor and predict air quality levels using AQI (CPCB) and Weather data.
- Implemented data preprocessing, feature engineering (lag features, seasonal trends), and model training using XGBoost.
- Designed a pipeline for real-time AQI prediction and actionable insights for urban environments.
- Integrated multiple datasets (CPCB + weather data) to improve prediction accuracy and robustness.

Multi-Objective Analysis of Spiking Neuron Models (Python)

[GitHub](#)

Jan. 2026
Research Project

- Built multi-objective evaluation framework for SNN models.
- Compared models on efficiency and biological realism.
- Simulated responses under varied stimulation protocols.
- Analyzed spike trains and membrane dynamics.

Spiking Neuron Simulations (Python, Brian2 / NEURON)

[GitHub](#)

Dec. 2025 - Present
Self-Directed Practice Repository

- Maintained a practice repository containing multiple implementations of spiking neuron models under varied conditions.
- Explored neuron behavior across different parameter regimes, input currents, and initial conditions.
- Used visualization of membrane potentials and spike trains to analyze model behavior.

Early-Stage Heart Attack Prediction (Python, ML)

[GitHub](#)

Oct. 2025
Machine Learning Project

- Built classification models for early heart attack prediction.
- Performed data preprocessing and feature engineering.
- Implemented CatBoost and XGBoost for improved accuracy.
- Optimized models using attention-based enhancements.

PUBLICATIONS

Abhishek Kumar, "Computational Neuroscience-Inspired Neuromorphic Computing: A Review of Models, Hardware Architectures, and Applications". (Manuscript submitted to a journal; [Presented at ICMSCI 2026](#))

CONFERENCE AND ACADEMIC EVENTS

Gwalior AI Summit' 26 [Certificate](#)

Prototype Presented

March 2026
ABV-IIITM Gwalior

- * Presented a prototype of **NEXUS**, a neuromorphic edge intelligence system, at the summit.
- * Participated in **AI Synergy 2026 Hackathon**, collaborating on innovative AI solutions.
- * Attended insightful lectures on neuromorphic computing, quantum computing, and AI for energy.
- * Engaged in a 2-day summit with industry professionals, researchers, and academic experts.

ICMSCI'26 [Certificate](#)

Paper Presented

Feb. 2026
PGDAV College, UoD

- * Presented a paper at International Conference on Mathematical Sciences and Computational Intelligence.
- * Topic: Computational Neuroscience-Inspired Neuromorphic Computing.
- * Analyzed TrueNorth, Loihi, SpiNNaker, and Neurogrid architectures.
- * Focused on energy-efficient AI and edge applications.

MATHSD'25 [Certificate](#)

Volunteering

Oct. 2025
ABV-IIITM Gwalior

- * Volunteered at MATHSD 2025, hosted by ABV-IIITM Gwalior.
- * Handled LaTeX typesetting and compilation of the official **Book of Abstracts**.
- * Assisted faculty and research teams in documentation and on-site coordination.
- * Gained insights on *Neuroscience-Inspired Scientific Machine Learning* through keynote sessions.

Samuchchayam 2025 **Certificate**

March 2025

Presented Paper

Deshbandhu college, UoD

- * Awarded first prize in the 'Datamatrix' paper presentation competition.
- * Recognized for excellence in research, analysis, and public speaking at a competitive academic event
- * Successfully developed and presented a compelling paper on a data-centric topic to a panel of faculty and peers.

InitMath 2024 **Certificate**

March 2024

Mathematics Camp

Uttarkashi (U.K)

- * Selected for and completed the 'Initiation into Mathematics (InitMath) 2024' training program.
- * Participated in an intensive workshop sponsored by the prestigious **National Board for Higher Mathematics (NBHM)**.
- * Engaged in a 6-day program focused on foundational concepts in advanced mathematics, organized by **MTTS Trust**.

TRAINING AND CERTIFICATIONS

Computational Neuroscience – University of Washington **Certificate**

Nov. 2025 - Jan. 2026

- * Strengthened understanding of the intersection between Neural biology and computation.
- * Instructors: Rajesh P. N. Rao, Adrienne Fairhall

Introduction to Cognitive Psychology and Neuropsychology - University of Cambridge **Certificate**

Jan. 2026 - Present

- * Learned theories, models and findings of cognitive functions, including memory, attention and perception.
- * Instructors: Dr. Giulia Mangiaracina

Linear Algebra for Machine Learning – LDRP Institute of Technology & Research **Certificate**

Dec. 2025

- * National Short-Term Training Program
- * Hands-on experience on PCA, SVD, and image processing.

ACHIEVEMENTS & AWARDS

National Yoga Championship Runner up **Certificate**

Sept. 2023

- * Championship held in Haridwar and organised by **World Fitness Federation of Yogasana Sports, India**

Certificate of Commendation by Chief Minister of Uttarakhand **Certificate**

June 2020

- * Award presented to students with outstanding achievements in academics.

SKILLS & INTERESTS

Languages: Python, MySQL, R

Tools: Power BI, Excel, Mathematica, MATLAB, Arduino IDE

Frameworks & Libraries: Numpy, Pandas, Matplotlib, Seaborn, Scikitlearn, PyTorch, snnTorch, NEURON

Domains: Computational Neuroscience, Machine Learning, Deep Learning, Neural Network

Others: Kaggle, Apache Kafka , Git, GitHub